

LESZNO, Poland

WMO: 124180

Lat: 51.8355N

Lon: 16.5347E

Elev: 93

StdP: 100.21

Time Zone: 1.00 (EUC)

Period: 94-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.5	-11.3	-16.6	0.9	-14.2	-13.7	1.2	-10.9	12.1	6.1	11.0	4.9	2.1	60	0.550

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	30.2	19.6	28.4	19.0	26.7	18.2	21.0	27.4	20.1	26.1	19.2	24.7	3.3	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	13.9	22.9	18.0	13.1	21.8	17.2	12.4	21.4	61.0	27.5	57.9	26.3	54.9	24.9	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(4) 9.6	8.2	7.2		-18.1	33.9	4.3	1.7	-21.3	35.1	-23.8	36.1	-26.3	37.0	-29.4		
(5)				-18.4	22.5	3.9	1.3	-21.3	23.5	-23.6	24.3	-25.8	25.0	-28.7	26.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	-1.3	0.3	3.5	8.9	13.5	16.5	19.1	18.4	13.8	9.2	4.5	0.1
	DBStd	8.18	5.38	5.03	4.26	4.11	3.75	3.38	3.33	3.05	3.20	3.93	3.72	4.94
	HDD10.0	1449	351	273	203	67	10	1	0	0	4	63	169	308
	HDD18.3	3564	609	506	459	283	156	76	31	37	140	283	417	566
	CDD10.0	1052	0	0	3	35	119	194	282	262	118	38	2	0
	CDD18.3	126	0	0	0	1	6	20	54	41	4	0	0	0
	CDH23.3	1409	0	0	0	17	92	254	607	388	49	2	0	0
	CDH26.7	389	0	0	0	2	13	62	208	97	7	0	0	0
(14) Wind	WSAvg	3.3	3.8	3.9	4.0	3.6	3.1	3.0	2.9	2.9	2.9	2.8	3.3	3.6
(15) Precipitation	PrecAvg	556	35	31	40	32	55	63	79	67	44	36	36	37
	PrecMax	731	67	67	84	84	104	139	195	177	107	100	83	84
	PrecMin	381	6	2	11	3	16	16	14	24	11	5	0	11
	PrecStd	85	16	15	19	18	21	32	44	32	25	23	17	17
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.5	13.0	17.7	25.7	28.6	31.2	33.6	31.5	27.6	22.3	15.0	11.6
		MCWB	8.9	9.1	10.8	15.6	18.9	20.6	20.3	19.9	18.3	16.9	12.2	9.7
	2%	DB	9.0	10.2	14.7	21.9	26.0	28.4	31.2	29.1	24.6	19.6	12.8	9.3
		MCWB	7.4	7.5	9.5	13.3	17.3	19.2	19.5	19.7	16.8	14.9	10.4	7.4
	5%	DB	7.1	8.6	12.3	18.9	23.6	26.3	28.9	27.3	22.4	17.5	11.5	7.7
		MCWB	5.7	6.4	8.4	12.0	16.0	17.9	19.2	18.5	16.0	13.9	9.7	6.3
	10%	DB	5.6	7.0	10.2	16.8	21.4	24.1	26.9	25.4	20.3	15.3	10.1	6.2
		MCWB	4.3	5.1	7.4	11.0	14.9	17.1	18.6	17.9	15.3	12.7	8.6	5.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.2	9.8	11.8	16.2	19.6	22.4	22.1	21.7	19.1	17.1	12.6	10.2
		MCDB	10.7	12.1	16.4	24.4	26.6	27.8	29.1	28.6	24.5	21.0	14.3	11.5
	2%	WB	7.5	8.1	10.4	14.4	18.1	20.2	21.0	20.6	17.9	15.5	11.1	7.7
		MCDB	8.9	9.8	13.7	20.3	24.2	26.7	28.0	26.4	23.2	18.9	12.3	8.8
	5%	WB	5.8	6.6	8.8	12.7	16.9	19.0	20.2	19.7	16.8	14.1	9.9	6.4
		MCDB	7.0	8.4	11.7	17.8	22.2	24.4	26.4	25.1	20.8	16.8	11.3	7.6
	10%	WB	4.4	5.2	7.5	11.5	15.7	17.8	19.3	18.8	15.8	13.0	8.6	5.1
		MCDB	5.5	6.9	10.0	15.9	20.3	22.4	25.1	23.5	19.4	15.0	10.0	6.1
(35) Mean Daily Temperature Range	5% DB	MDBR	4.9	6.2	8.0	10.6	10.9	11.1	11.0	10.9	9.4	8.4	6.0	4.9
		MCDBR	6.0	8.5	12.7	14.7	14.7	15.1	15.0	14.3	13.6	11.4	7.9	5.9
		MCWBR	5.0	6.1	8.0	7.8	7.3	7.1	5.6	6.0	6.8	7.1	5.9	4.8
	5% WB	MCDBR	5.8	7.9	11.2	13.3	12.8	13.3	12.5	12.1	11.8	10.7	7.6	5.7
		MCWBR	5.0	6.1	7.5	7.4	7.0	6.9	5.4	5.7	6.4	7.0	5.9	4.8
(40) Clear-Sky Solar Irradiance	taub		0.318	0.343	0.381	0.425	0.419	0.416	0.439	0.432	0.391	0.368	0.344	0.312
	taud		2.320	2.303	2.228	2.159	2.226	2.264	2.212	2.244	2.328	2.358	2.361	2.346
	Ebn at Noon		677	765	801	808	835	840	810	794	785	726	639	620
	Edh at Noon		64	86	112	135	132	128	133	123	101	81	62	54
(44) All-Sky Solar Radiation	RadAvg		0.72	1.41	2.54	4.16	5.12	5.38	5.04	4.47	3.21	1.80	0.83	0.53
	RadStd		0.12	0.26	0.33	0.55	0.58	0.49	0.59	0.40	0.41	0.24	0.12	0.06

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)	+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+57	N/A

Nomenclature: See separate page